

Unit 105 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 57.4 ac, BAF 20 variable-radius cruise on a 300 ft grid, 29 plots, 322 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	222 sq ft/ac (1 SE of mean, BA 2.1 %)	Type	Sierran mixed conifer
Trees per acre	71	CWHR size / density	4 / D (cover 93 %)
QMD	24.0 in	SDI	274 (48 % of max 565)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	221 sq ft/ac 24.8 in	BA QMD	1 sq ft/ac 7.2 in
TPA	66	TPA	5
Volume	39.0 MBF/ac (7775 cu ft)	Volume	0.6 green tons/ac (25 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~161 sq ft/ac BA	Sampling error (volume)	8.7 % at 95 % (CV 23 %, t = 2.05, n = 29)
Species (BA)	PP 72, WF 60, DF 34, IC 32, SP 23	Stratum standard	40 % of volume - MEETS; plots for standard: 2
		Sampling error (basal area)	4.3 % at 95 % (CV 11 %); secondary, not the standard
		Plot count	29 vs 20-plot local practice, not a handbook standard: ok

Cubic volume = total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium) net of defect; Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006); green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber, so applied to total-stem CVTS it overstates sawlog volume; merchantable cubic volume would be the consistent pairing. Sampling error at 95 % confidence with Student's t on plot net cubic volume per acre, the quantity the handbook standards apply to (FSH 2409.12 ch. 40, 41.1(5)(a) and (b)); stratum standard 40 % of volume for tree-measurement sales; the basal-area error is secondary. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables exclude records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	1	3.2	0.7	12	0.1
8-11.9	5	6.1	3.4	65	0.3
12-15.9	16	9.6	11.0	266	1.3
16-19.9	36	13.9	24.8	675	3.4
20-23.9	50	13.2	34.5	1039	5.2
24-27.9	46	8.5	31.7	1022	5.1
28-31.9	48	6.8	33.1	1174	5.9
32-35.9	43	4.8	29.7	1078	5.4
36-39.9	18	1.6	12.4	521	2.6
40-43.9	17	1.2	11.7	475	2.4
44-47.9	10	0.6	6.9	312	1.6
48-51.9	11	0.6	7.6	381	1.9
52-55.9	4	0.2	2.8	151	0.8
56-59.9	6	0.2	4.1	215	1.1
60-63.9	11	0.4	7.6	415	2.1
All	322	70.9	222.1	7800	39.2

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	105	21.5	72.4	24.9	2883	14.5
WF	87	22.1	60.0	22.3	2011	10.1
DF	50	12.0	34.5	23.0	1114	5.6
IC	46	7.6	31.7	27.6	928	4.7
SP	34	7.7	23.4	23.6	864	4.3
All	322	70.9	222.1	24.0	7800	39.2

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 105: plots on the 300 ft grid (orange = QA flag)

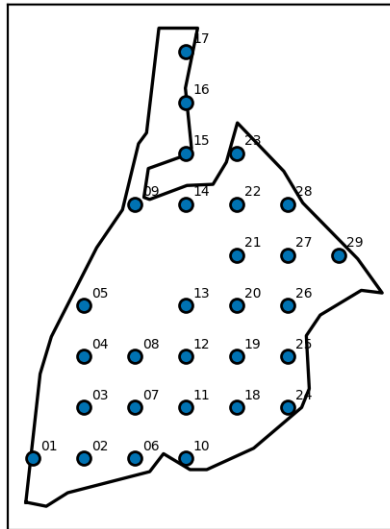


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
105-01	11	220	11	ok
105-02	12	240	18	ok
105-03	12	240	14	ok
105-04	14	280	32	ok
105-05	10	200	20	ok
105-06	10	200	16	ok
105-07	11	220	6	ok
105-08	12	240	26	ok
105-09	11	220	29	ok
105-10	13	260	22	ok
105-11	11	220	20	ok
105-12	10	200	20	ok
105-13	10	200	22	ok
105-14	10	200	19	ok
105-15	9	180	33	ok
105-16	10	200	36	ok
105-17	11	220	28	ok
105-18	11	220	12	ok
105-19	8	160	22	ok
105-20	12	240	28	ok
105-21	11	220	26	ok
105-22	12	240	18	ok
105-23	13	260	26	ok
105-24	12	240	15	ok
105-25	10	200	18	ok
105-26	11	220	28	ok
105-27	12	240	13	ok
105-28	12	240	19	ok
105-29	11	220	16	ok